**UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES (UITS)**
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**
DHAKA, BANGLADESH

**CSE 416: Software Project Design and Development (Lab)**

# SOFTWARE TEST REPORT

## The Round Table Exchange (RTE): A Multi-Party Skill Barter Network with Graph-Based Cycle Matching

**Submitted By:**
Ahmmad Abdali Khan (ID: 0432320005101118)
Sumaia Bintey Ismail (ID: 0432320005101103)
Alin (ID: 0432320005101131)
*Batch: CSE-54, Section: 7A*

**Submitted To:**
**Dr. Mahfida Amjad Dipa**
Assistant Professor
Department of Computer Science and Engineering
University of Information Technology and Sciences (UITS)

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# Document Control

| Parameter | Configuration Reference |
|---|---|
| Repository | https://github.com/Veil-Proxima/RTE |
| Version Tested | v1.0.1-test (commit 141fe65) |
| Live Client Site | https://veil-proxima.github.io/RTE (Local API: http://127.0.0.1:8000) |
| Report Version | 1.0 (Baseline quality tier) |
| Prepared By | Ahammed Abdali Khan, Sumaia Bintey Ismail, Alin |
| Reviewed By | Ahammed Abdali Khan |
| Target Quality Tier | Baseline (ISO/IEC/IEEE 29119-3 & ISO/IEC 25010 compliant) |

## Document Revision History

| Version | Date | Summary of Verification Changes |
|---|---|---|
| 1.0 | 02 Nov 2026 | **First comprehensive baseline report: 78 automated test cases, unit + integration suites, RTM traceability, 98% core backend coverage, NFR performance benchmarks, and defect log.** |
| 1.1 | 05 Nov 2026 | Planned for Exp 10: Extended OWASP ZAP automated scanner, SonarCloud static analysis gates, and multi-browser Playwright matrix. |

# 1. Introduction & Overview

**1.1 Purpose:** This report details the verification of RoundTable Exchange (RTE) v1.0.1-test: test scope, methodology, automated test results, code coverage telemetry, defect analysis, and release readiness based on ISO/IEC/IEEE 29119-3 and ISO/IEC 25010.

**1.2 Product Overview:** RTE is a circular multi-party skill barter engine that eliminates the classic 'Double Coincidence of Wants' barrier by discovering closed barter cycles of length k in {2, 3, 4} using a Bounded DFS multigraph engine. Discovered cycles are evaluated via a composite function balancing Haversine geographic proximity with weekly availability schedules: S(C) = alpha * S_geo(C) + (1 - alpha) * S_sched(C). An atomic state machine coordinates unanimous single-veto consensus.

## 1.3 Scope of Testing

| Subsystem / Functional Area | SRS Requirements | Testing Levels |
|---|---|---|
| User onboarding & Dhaka coordinate profiles | FR-01 to FR-05, FR-24 | Unit, Integration |
| Skill taxonomy & multi-listing management | FR-06 to FR-10 | Unit, Integration |
| Directed Multigraph construction & edges | FR-11 to FR-14 | Unit, Integration |
| Bounded DFS cycle discovery (k in [2, 4]) | FR-15 to FR-18 | Unit, Integration, Algorithmic |
| Rotational cycle deduplication & DAG safety | FR-19 to FR-21 | Unit, Algorithmic |
| Haversine geodesic distance ranking | FR-22 to FR-23 | Unit, Integration |
| Weekly availability grid set intersection | FR-24 to FR-26 | Unit, Integration |
| Dynamic composite scoring & real-time alpha | FR-27 to FR-28 | Unit, Client Engine |
| Atomic 5-stage proposal state machine | FR-29 to FR-33 | Unit, Integration |
| Synthetic population generator & scaling | FR-34 to FR-36 | Performance, Load |
| REST API gateway contract validation | FR-37 to FR-40 | Integration, Security |
| Client-side standalone JavaScript engine | FR-41 to FR-43 | Client Unit, Parity |
| Responsive ForceAtlas2 physics canvas | FR-44 to FR-46 | UI, Cross-Browser |
| Performance, Security, and Code Coverage | NFR-01 to NFR-16 | Perf, Sec, Coverage |

## 2. Test Environment & Strategy

| Component | Specification & Tooling Configuration |
|---|---|
| Host Operating System | Windows 11 Enterprise (local test harness); Linux x86_64 (CI container) |
| Language Runtimes | Python 3.15.0 (64-bit AMD64), Node.js v24.22.1 |
| Backend Frameworks | FastAPI 0.115, Starlette, Pydantic v2.10, Uvicorn 0.34, NetworkX 3.4 |
| Test Runners & Libraries | pytest 9.1.1, pytest-cov 6.0, coverage.py 7.6, anyio 4.15, Node test runner |
| Test Store | In-memory Multigraph and clean dictionary store; fresh instances on every test run |
| Browsers Tested | Google Chrome 131, Microsoft Edge 131 desktop, responsive mobile viewports |

### 2.1 Exit Criteria Evaluation (Baseline Quality Tier)

| Evaluation Criterion | Target Metric | Achieved Result | Status |
|---|---|---|---|
| Functional Requirements Coverage | **100%** | **100% (43 / 43 FRs covered)** | **Pass** |
| **P1 Critical Test Pass Rate** | **100%** | **100% (52 / 52 P1 cases passed)** | **Pass** |
| Core Backend Code Coverage | >= 85% | **98% (523 / 536 statements)** | **Pass** |
| Open Severity 1 (S1) Defects | 0 | 0 (DEF-01 resolved in 141fe65) | **Pass** |
| Open Severity 2 (S2) Defects | 0 | 0 | **Pass** |
| Dual-Engine Mathematical Parity | **100%** | **100% Identical Output (94.7%)** | **Pass** |

## 3. Requirements Traceability Matrix (RTM)

| Req ID | Functional Requirement Description | Test Case IDs | Status |
|---|---|---|---|
| FR-01 | User registration with Dhaka coordinate profiles | TC-01, TC-02, TC-03 | **Pass** |
| FR-02 | User listing query with metadata retrieval | TC-04, TC-05 | **Pass** |
| FR-03 | User deletion with cascading store cleanup | TC-06, TC-07 | **Pass** |
| FR-04 | Input validation on user profile coordinates | TC-08, TC-09 | **Pass** |
| FR-05 | HTTP 404 response on non-existent entities | TC-10 | **Pass** |
| FR-06 | Skill offer listing registration & category | TC-11, TC-12 | **Pass** |
| FR-07 | Skill want request registration & validation | TC-13, TC-14 | **Pass** |
| FR-08 | Prevention of identical offer/want on user | TC-15 | **Pass** |
| FR-09 | Skill taxonomy & location metadata endpoint | TC-16, TC-17 | **Pass** |
| FR-10 | Case-insensitive trimmed skill matching | TC-18, TC-19 | **Pass** |
| FR-11 | MultiDiGraph node synthesis from users | TC-20, TC-21 | **Pass** |
| FR-12 | Directed edge insertion matching offer->want | TC-22, TC-23 | **Pass** |
| FR-13 | Edge attributes storing transferred skill | TC-24 | **Pass** |
| FR-14 | Multi-edge support between users | TC-25 | **Pass** |
| FR-15 | Bounded DFS 3-party loop detection (k=3) | TC-26, TC-27 | **Pass** |
| FR-16 | Bounded DFS 4-party loop detection (k=4) | TC-28, TC-29 | **Pass** |
| FR-17 | Strict cycle length bounding (min_k, max_k) | TC-30, TC-31 | **Pass** |
| FR-18 | Rotational canonical cycle deduplication | TC-32, TC-33 | **Pass** |
| FR-19 | Zero false-positive cycles on DAG topologies | TC-34, TC-35 | **Pass** |
| FR-20 | Early DFS recursion pruning (> max_k) | TC-36 | **Pass** |
| FR-21 | Cycle discovery limit safety cap (100 max) | TC-37 | **Pass** |
| FR-22 | Haversine geodesic distance calculation | TC-38, TC-39 | **Pass** |
| FR-23 | Proximity score exponential decay modeling | TC-40, TC-41 | **Pass** |
| FR-24 | Weekly availability grid modeling (7x24) | TC-42 | **Pass** |
| FR-25 | Mutual schedule overlap set intersection | TC-43, TC-44 | **Pass** |
| FR-26 | Normalized schedule overlap scoring (hrs/6) | TC-45 | **Pass** |
| FR-27 | Parametric composite scoring with alpha | TC-46, TC-47 | **Pass** |
| FR-28 | Descending composite score sorting | TC-48 | **Pass** |
| FR-29 | Trade proposal initialization & UUID | TC-49, TC-50 | **Pass** |
| FR-30 | Multi-party consensus tracking (Pending/Accepted) | TC-51, TC-52 | **Pass** |
| FR-31 | Unanimous acceptance transition to Locked | TC-53, TC-54 | **Pass** |
| FR-32 | Single-veto immediate transition to Rejected | TC-55, TC-56 | **Pass** |
| FR-33 | Response rejection on invalid proposal ID | TC-57, TC-58 | **Pass** |
| FR-34 | Synthetic Dhaka population generator | TC-59, TC-60 | **Pass** |
| FR-35 | Empirical scalability benchmark runner | TC-61, TC-62 | **Pass** |
| FR-36 | Graph build vs DFS cycle search telemetry | TC-63 | **Pass** |
| FR-37 | REST API /api/meta endpoint | TC-64 | **Pass** |
| FR-38 | REST API /api/match/run endpoint | TC-65, TC-66 | **Pass** |
| FR-39 | REST API /api/match/cycles cached query | TC-67 | **Pass** |
| FR-40 | REST API /api/reset state restore endpoint | TC-68 | **Pass** |
| FR-41 | Standalone client JavaScript DFS engine | TC-69, TC-70 | **Pass** |
| FR-42 | JavaScript Haversine & schedule score parity | TC-71, TC-72 | **Pass** |
| FR-43 | Real-time client alpha slider re-scoring | TC-73 | **Pass** |
| NFR-01 | Algorithmic matching < 50ms for N <= 100 | PT-01 | **Pass** |
| NFR-02 | REST API response time < 100ms under load | PT-02, PT-03 | **Pass** |
| NFR-04 | Dual-engine mathematical score parity (94.7%) | TC-74 | **Pass** |

# 4. Execution Results & Code Coverage Telemetry

## 4.1 Automated Test Execution Summary

| Testing Level | Total Runs | Passed | Failed | Pass Rate (%) |
|---|---|---|---|---|
| Algorithmic Core Unit Tests | 35 | 35 | 0 | 100.0% |
| REST API Gateway Integration Tests | 29 | 29 | 0 | 100.0% |
| Client-Side Standalone Engine Tests | 14 | 14 | 0 | 100.0% |
| Total Automated Test Suite | 78 | 78 | 0 | 100.0% |

## 4.2 Backend Statement Code Coverage (98% Achieved)

| Module File | Statements | Missing | Coverage (%) | Verified Functional Scope |
|---|---|---|---|---|
| backend/app.py | 140 | 0 | **100%** | FastAPI REST routes, schema validation & static mounts |
| backend/models.py | 52 | 0 | **100%** | Pydantic domain entities (User, SkillListing, TradeCycle) |
| backend/synthetic_data.py | 54 | 0 | **100%** | Dhaka demographic generator & scaling benchmarks |
| backend/test_api.py | 74 | 0 | **100%** | Full integration suite for HTTP GET/POST/DELETE paths |
| backend/test_engine.py | 70 | 0 | **100%** | Algorithmic DFS bounds, DAG acyclic checks, consensus |
| backend/graph_engine.py | 109 | 7 | 94% | MultiDiGraph, bounded DFS cycle discovery & scoring |
| backend/workflow.py | 37 | 6 | 84% | Atomic multi-party proposal state transitions & voting |
| TOTAL SYSTEM CORE | 536 | 13 | **98%** | Full System Automated Verification (Target >= 85%) |

## 4.3 Scalability & Performance Benchmarks

| Scale (N) | Edges (M) | Graph Build | Cycle DFS | Total Runtime | Loops Found (k in [3,4]) |
|---|---|---|---|---|---|
| 10 users | 24 edges | 0.21 ms | 0.45 ms | 0.66 ms | 2 cycles |
| 25 users | 86 edges | 0.62 ms | 1.84 ms | 2.46 ms | 7 cycles |
| 50 users | 210 edges | 1.45 ms | 8.20 ms | 9.65 ms | 19 cycles |
| 100 users | 534 edges | 3.80 ms | 38.10 ms | 41.90 ms | 47 cycles |
| 200 users | 1,420 edges | 9.10 ms | 287.30 ms | 296.40 ms | 162 cycles |
| 247 users | 1,840 edges | 12.40 ms | 538.60 ms | 551.00 ms | 241 cycles (capped at 100) |

## 5. Defect Log & Main Bug Fixes

| ID | Req Ref | Problem Description | Sev | Status | Resolution Reference |
|---|---|---|---|---|---|
| DEF-01 | FR-42 | Client cycle scores showed static 80.0% due to dummy constants. | S1 | **Closed** | Commit 141fe65 (Haversine & schedule port) |
| DEF-02 | FR-18 | Duplicate rotational cycles returned when start node was not lowest ID. | S2 | **Closed** | Canonical rotational slicing in DFS |
| DEF-03 | FR-32 | Proposal remained Accepting after a member rejected. | S2 | **Closed** | Single-veto short-circuit to Rejected |
| DEF-04 | FR-04 | User creation accepted coordinates outside Bangladesh. | S2 | **Closed** | Pydantic coordinate validator |
| DEF-05 | FR-43 | Alpha slider updated label but did not re-rank cycles dynamically. | S3 | **Closed** | Real-time input event re-sorting |
| DEF-06 | FR-21 | Dense synthetic graphs caused memory spikes in DFS. | S3 | **Closed** | Depth pruning & 100 cycle limit |
| DEF-07 | FR-10 | Skill matching failed on mixed case (e.g. python vs Python). | S3 | **Closed** | Lowercase string normalization |
| DEF-08 | FR-44 | Vis.js network nodes collapsed on small viewports. | S4 | **Closed** | ForceAtlas2 springLength tuning |

### 5.1 Root Cause Analysis of Primary Bug (DEF-01)

In frontend/js/app.js, findClientCycles() contained static placeholder constants (proxScore = 0.85, overlapScore = 0.75). With alpha = 0.5, (0.5 * 0.85) + (0.5 * 0.75) = 0.80 (80.0%) across every cycle. We resolved this by implementing haversineDistanceKm and calculateScheduleOverlapHours in JavaScript, matching the Python backend exactly (Cycle 1: 94.7%, Cycle 2: 69.8%).

## 6. Individual Contributions & Sign-off

| Member Name | Student ID | Assigned Testing Role & Responsibilities | Found | Fixed |
|---|---|---|---|---|
| Ahammed Abdali Khan | 0432320005101118 | Bounded DFS verification, Haversine math, client scoring parity, test architecture | 4 | 4 |
| Sumaia Bintey Ismail | 0432320005101103 | Consensus testing, schedule overlap grid, integration test harness, documentation | 3 | 3 |
| Alin | 0432320005101131 | Synthetic benchmarks, REST API contracts, negative security tests, QA review | 1 | 1 |

**Release Recommendation:** RECOMMENDATION: PROCEED WITH RELEASE (v1.0.1-test). The system demonstrates exceptional algorithmic stability, 100% automated test pass rate across 78 test cases, 98% core backend statement coverage, and zero open critical defects.

**Declaration:** We declare that this report accurately describes the comprehensive software testing conducted on RoundTable Exchange (RTE) version v1.0.1-test (commit 141fe65). All test cases, coverage telemetry, and performance benchmarks are authentic and reproducible directly from the repository.